

CASE: Conflict-assessed Knowledge-sensitive Neuron Tuning for Lifelong Model Editing

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Abstract

Large Language Models (LLMs) inevitably encounter factual hallucinations and knowledge obsolescence, necessitating lifelong knowledge editing to sustain reliability and factual advancement. While mainstream lifelong editing paradigms aim to alleviate knowledge forgetting through allocating and updating isolated parameter subspaces, they often overlook conflict assessment among distinct editing processes, leading to unjustified subspace allocation and indiscriminate neuron tuning. To address these issues, we propose the Conflict-Assessed Sensitive Editing (CASE) framework, which integrates a Conflict-Assessed Editing Allocation (CAA) module and a Knowledge-sensitive Neuron Tuning (KNT) strategy. The CAA module quantitatively assesses editing conflicts to enable justified subspace allocation, thereby reducing globally significant conflicts and routing errors. The KNT strategy adaptively identifies and

tunes knowledge-sensitive neurons through a calibrated sensitivity threshold, effectively eliminating local conflicts and enhancing subspace stability. Extensive experiments on standard lifelong editing benchmarks demonstrate that CASE achieves state-of-the-art performance, improving average editing accuracy by nearly 10% after 1,000 sequential edits. Overall, CASE substantially mitigates editing conflicts and enhances knowledge retention, offering a scalable and conflict-resilient solution for lifelong model editing.

CCS Concepts

• **Computing methodologies** → **Artificial intelligence; Knowledge representation and reasoning.**

Keywords

Large Language Models, Knowledge Editing, Lifelong Model Editing, Conflict Assessment

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1 Introduction

Model editing has emerged as an effective paradigm for correcting factual inaccuracies in large language models (LLMs) [1, 26], including hallucinations, spurious reasoning, and knowledge obsolescence [10, 18, 27]. Model editing seeks to precisely update targeted knowledge while preserving the model’s unrelated information [6, 21, 22]. However, since real-world knowledge evolves in a continuous and asynchronous manner, it is essential to enable continual model updates to preserve the temporal relevance of an LLM’s knowledge base [7, 27]. To address this issue, Hartvigsen et al. [7] introduced the Lifelong Model Editing task, which aims to allow LLMs to continuously assimilate new knowledge while retaining all previously incorporated edits.

The central challenge in lifelong model editing lies in the fact that different edits may induce conflicting parameter updates, ultimately leading to catastrophic forgetting of previously integrated knowledge [6, 11]. To alleviate this issue, mainstream studies have primarily focused on allocating distinct parameter subspaces for different edits to separate potential conflicts, while employing routing mechanisms during inference to select the most relevant subspaces. Specifically, GRACE [7] preserves newly incorporated knowledge through additional key-value pairs and designs a discrete codebook of edits to retrieve and replace corresponding layer representations during inference. WISE [27] allocates a side memory after a fixed number of edits and trains an inference-time routing mechanism to select the appropriate side-memory region. Similarly, MELO [31] and LEMOE [29] assign independent auxiliary experts to each edit and train a semantic-similarity-based gating module that selectively activates relevant knowledge subspaces during inference.

Nevertheless, as illustrated in Figure 1, existing approaches lack a quantitative assessment of update conflicts among multiple edits, resulting in suboptimal subspace allocation and inference routing, as well as limited editing retention: **(1) Unjustified Subspace Allocation.** Current methods assign parameter subspaces to a fixed number of sequential edits without considering the quantitative degree of conflict among them, resulting in either over-conflicted or over-fragmented subspaces. Consequently, conflicting edits within the same subspace cause over-conflicted updates and catastrophic forgetting, while over-fragmented subspaces introduced to mitigate such conflicts results in parameter redundancy and increased inference routing complexity. **(2) Indiscriminate Dense Tuning.** Within each subspace, existing methods typically apply uniform dense updates without accounting for neuron-level updating sensitivity across editing instances. Distinct knowledge edits typically rely on different subsets of highly sensitive neurons. Such indiscriminate dense parameter updates disperse gradients into insensitive neurons, amplifying residual local conflicts of different instances and ultimately causing cumulative catastrophic forgetting over sequential edits.

In this paper, we propose the Conflict-Assessed Sensitive Editing (CASE) framework, which quantitatively assesses the editing conflicts to guide justified subspace allocation and effective editing performance. The CASE framework comprises two core components: a Conflict-Assessed Editing Allocation (CAA) module and a Knowledge-sensitive Neuron Tuning (KNT) strategy. The CAA module quantitatively assesses editing conflicts among knowledge

instances and existing editing memory to enable justified subspace allocation. It assigns largely consistent instances to shared subspaces while isolating highly conflicting ones, thereby eliminating over-conflicted updates and preventing over-fragmented subspaces to reduce routing difficulty. The KNT strategy further adaptively selects and tunes knowledge-sensitive neurons within each subspace through a calibrated threshold, focusing each edit on its most responsive neurons to suppress local conflicts and preserve subspace stability. By integrating the CAA and KNT components, CASE achieves justified subspace allocation and neuron-level adaptive editing. This joint design effectively mitigates knowledge forgetting and routing inaccuracies while eliminating localized neuron-level conflicts, thereby enabling stable lifelong editing.

Our contributions are as follows:

- We propose the Conflict-Assessed Sensitive Editing (CASE) framework, the first to explicitly assess inter-edit conflicts in lifelong model editing, thereby providing an effective and justified solution to subspace-based lifelong model editing.
- We develop a Conflict-Assessed Editing Allocation (CAA) module that quantitatively assesses editing conflicts and justifiably allocates each edit to a consistent subspace, addressing the knowledge forgetting and routing inaccuracies caused by over-conflicted or over-fragmented subspaces.
- We introduce a Knowledge-sensitive Neuron Tuning (KNT) strategy that adaptively selects and tunes knowledge-sensitive neurons within each subspace, substantially mitigating localized conflicts and enhancing the knowledge retention.
- We conduct extensive experiments on a widely used lifelong editing benchmark, showing that CASE consistently outperforms state-of-the-art methods, achieving nearly a 10% accuracy gain over the second-best approach on 1000 sequential edits.

2 Related Work

Our work aims to address the problem of knowledge forgetting caused by update conflicts in lifelong model editing, which is primarily related to the research of model editing and continual learning.

2.1 Model Editing

Model editing aims to update targeted knowledge while preserving the other irrelevant knowledge [6, 21, 22], which can be divided into two categories: single-batch editing and lifelong editing.

Single batch editing. Early model editing methods focus on single batch edit scenarios. ROME [21] locates editable layers via causal mediation analysis and updates parameters through the key-value mechanism. MEMIT [22] extends this paradigm to simultaneous batch editing of multiple samples. COMEBA-HK [14] refines it by defining the Local Editing Scope and enabling sequential editing. MEND [23] trains a hypernetwork to decompose fine-tuning gradients into generalizable update directions that preserve performance on unrelated inputs.

Lifelong editing. To tackle dynamically evolving real-world knowledge, Hartvigsen et al. [7] proposed Lifelong Model Editing, enabling LLMs to incrementally integrate new knowledge while preserving consistency with prior edits. Existing lifelong editing methods mainly use parameter expansion or projection to

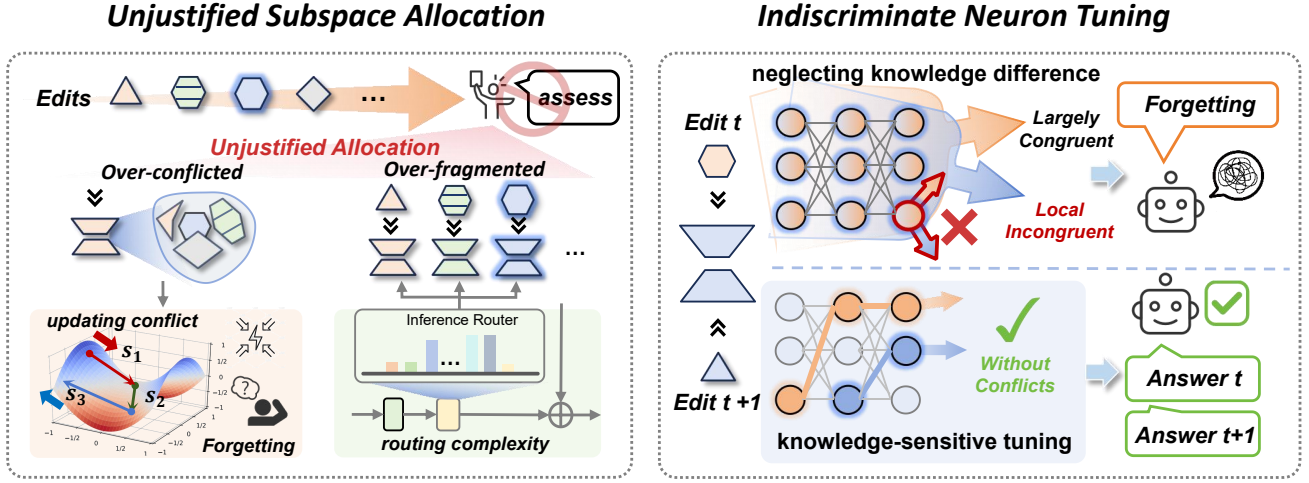


Figure 1: Current lifelong editing approaches lack a quantitative evaluation of update–editing conflicts and exhibit two major deficiencies: unjustified subspace allocation and indiscriminate dense tuning.

allocate parameter subspaces for different edits, preserving prior knowledge integrity. Representative parameter projection method AlphaEdit [6] constrains updates to the null space of prior knowledge via a pre-trained projection matrix to reduce conflicting updates, but it relies on pre-training data distribution consistency, failing to fully eliminate practical editing conflicts. Parameter expansion methods (e.g., WISE [27]) allocate and merge side memory after fixed edits to mitigate update conflicts; similarly, MELO [31] and LEMOE [29] assign independent auxiliary experts to each edit and design a routing mechanism to selectively activate relevant knowledge subspaces during inference.

Nevertheless, these approaches lack a quantitative assessment of update conflicts across different edits when allocating and updating parameter subspaces, making it challenging to effectively eliminate conflicts and mitigate the resulting knowledge forgetting in lifelong model editing. In contrast, our proposed method is the first to explicitly assess inter-edit conflicts in lifelong model editing and to leverage these assessments for strategic subspace allocation and adaptive updating, thereby providing an effective and principled solution for conflict-free lifelong editing.

2.2 Continual Learning

Continual learning [25] addresses the catastrophic forgetting problem encountered when deep learning models acquire new knowledge [4]. Contemporary research investigates various approaches to mitigate forgetting while enabling effective knowledge accumulation. A common strategy is continual fine-tuning, in which LLMs are incrementally updated as new data becomes available; a comprehensive study of such methods is provided by [15]. However, it has been observed that conventional continual learning regularization techniques—such as Elastic Weight Consolidation [12], Experience Replay [24], and Maximally Interfered Replay [2]—although effective in preserving previous knowledge, frequently result in rapid degradation of performance on earlier tasks. Moreover, prior continual learning settings typically involve a limited number of tasks, often fewer than ten, whereas lifelong model editing requires handling hundreds or even thousands of editing steps, thereby posing greater challenges for long-term knowledge retention.

3 Method

Previous lifelong model editing approaches lack a quantitative evaluation of conflicts among editing instances, leading to unjustified subspace assignment and indiscriminate dense tuning. To overcome these limitations, we propose the Conflict-Assessed Sensitive Editing (CASE) framework for stable lifelong model editing. By justifiably allocating parameter subspaces based on quantitative conflicts assessment, while adaptively identifying and tuning knowledge-sensitive neurons, CASE effectively mitigates editing conflicts and inference routing inconsistencies, thereby achieving superior editing stability and precision. The framework comprises two core components: a Conflict-Assessed Editing Allocation (CAA) module and a Knowledge-Sensitive Neuron Tuning (KNT) strategy, which are detailed in Section 3.2 and Section 3.3, respectively.

3.1 Preliminary

Lifelong model editing task [7, 27] iteratively modifies the LLM $\mathcal{F}_0$ through a sequence of edits, enforcing consistency between generated and desired outputs for specified queries, without degrading performance on unmodified tasks. A t -step lifelong editing dataset is formulated as a sequential collection of targeted knowledge samples: $\mathcal{D}_e = \{s_1, \dots, s_t\} = \{(x_1, y_1), \dots, (x_t, y_t)\}$, where each pair $s_t = (x_t, y_t)$ corresponds to an edit operation at step t .

Several studies [7, 27, 31] have proposed lifelong editing approaches by augmenting specific layers with additional parameters $\mathbf{W}$, thereby facilitating the memorization of new knowledge during incremental updates. These approaches typically replicate the model’s original weights to edit targeted knowledge within a certain step range [27], or alternatively, they create new low-rank subspaces [9] for each edit. The editing process is typically optimized through cross-entropy minimization:

$$\mathbf{W}_t \leftarrow \mathbf{W}_{t-1} - \eta \cdot (\nabla \mathcal{L}_{CE}(y_t, \mathcal{F}(x_t; \mathbf{W}_{t-1}))) \quad (1)$$

where η is the learning rate, $\mathbf{W}_{t-1}$ denotes additional parameters of the model after step $t - 1$.

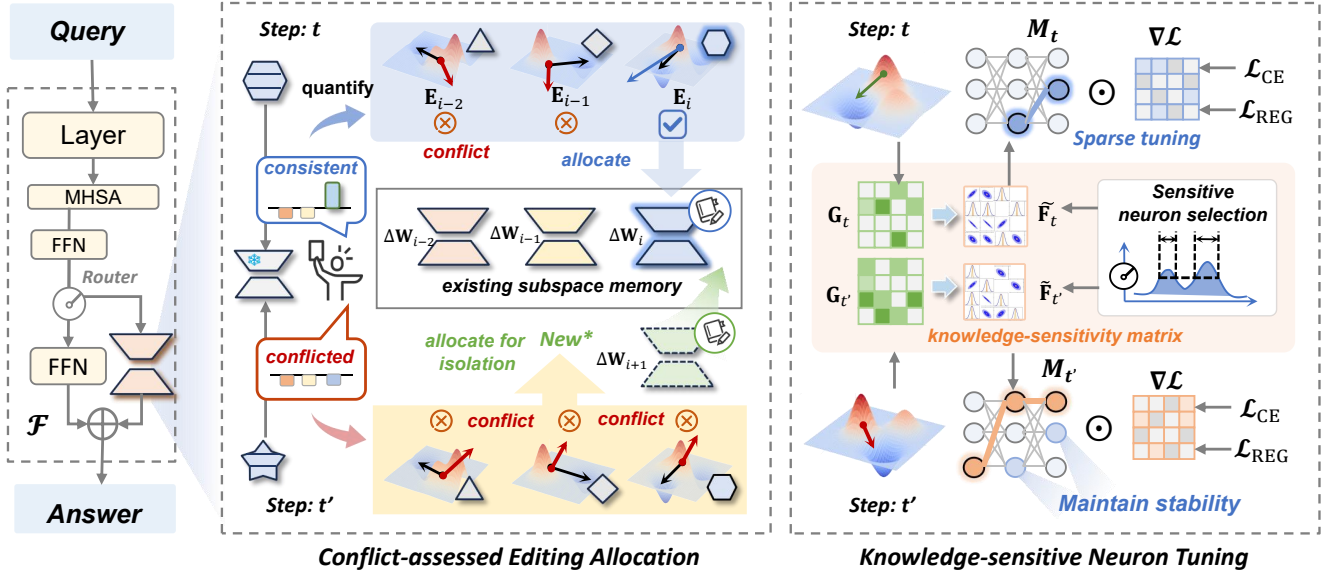


Figure 2: Overview of the CASE framework. The CAA module quantitatively assesses editing conflicts to enable justified subspace allocation, and the KNT strategy adaptively identifies and tunes knowledge-sensitive neurons through a sensitivity threshold, effectively enhancing subspace editing performance.

3.2 Conflict-Assessed Editing Allocation

Existing approaches allocate parameter subspaces in an unjustified manner without assessing editing conflicts, leading to either over-conflicted or over-fragmented subspaces. Therefore, we introduce the Conflict-Assessed editing Allocation (CAA) module, which performs quantitative conflict assessment to enable justified subspace allocation. CAA assesses the editing conflicts between each editing instance and existing subspaces, allocating largely consistent knowledge into shared subspaces while highly conflicting knowledge to newly instantiated subspaces for isolation. This adaptive mechanism mitigates editing conflict and inference routing difficulties caused by unjustified allocation.

To effectively allocate conflict-isolated and justified subspaces for targeted knowledge instances, we first quantify the editing congruency between the knowledge sample s_t and existing parameter subspaces $\mathcal{W}_{t-1} = \{\Delta W_{t-1}^1, \dots, \Delta W_{t-1}^n\}$, to assess the potential conflict of diverse knowledge within lifelong editing.

To this end, we systematically characterize the accumulated editing trajectories within existing subspaces and the potential editing difficulty of current targeted knowledge. Each subspace is initialized as the frozen LLM weight ΔW_0^0 and will be updated only if it is allocated for the current targeted knowledge. To precisely characterize the editing trajectories of subspaces, we assess the cumulative editing influence of the subspace through the deviation of each subspace's parameters from ΔW_0^0 :

$$E_{t-1}^i = \Delta W_0^0 - \Delta W_{t-1}^i \quad (2)$$

The editing difficulty for target knowledge $s_t = (x_t, y_t)$ can be quantified through the gradient matrix G_t of the original LLM's loss function. A high-norm gradient directly reflects substantial discrepancy between the model's current predictions and desired knowledge outputs, indicating significant conceptual discordance

that necessitates more intensive parameter adjustments:

$$G_t = -\nabla \mathcal{L}_{CE} \left(y_t, \mathcal{F} \left(x_t; \Delta W_{t-1}^0 \right) \right) \quad (3)$$

where $\mathcal{L}_{CE}$ denotes the cross-entropy editing loss. Therefore, the editing congruency between knowledge instance s_t and each subspace ΔW_{t-1}^i can be quantified through similarity of the editing gradient G_t and the subspace deviation E_{t-1}^i :

$$c_t^i = \cos \left(G_t, E_{t-1}^i \right) \quad (4)$$

The quantified congruency metric enables explicit conflict assessment: when $c_t^i \geq 0$, sample s_t exhibits geometric compatibility with subspace ΔW_{t-1}^i (denoting conflict absence). Values $c_t^i < 0$ trigger conflict identification, necessitating isolation of s_t from ΔW_{t-1}^i to preserve prior integrity.

To allocate most consistent subspace for knowledge sample s_t , we allocate the subspace exhibiting the highest editing congruency among all subspaces with minimal conflict assessed for editing. The subspace allocation operator ΔW_t^* is formally defined as:

$$\Delta W_t^* = \begin{cases} \arg \max_{\Delta W^i \in \mathcal{W}_{t-1}} \{c_t^i \mid c_t^i \geq 0\}, & \text{if } \max_i c_t^i \geq 0 \\ \Delta W_{t-1}^{n+1} \leftarrow \Delta W_0^0, & \text{otherwise} \end{cases} \quad (5)$$

$$\mathcal{W}_t = \mathcal{W}_{t-1} \cup \Delta W_t^* \quad (6)$$

Therefore, we could obtain the allocated subspace ΔW_t^* for the t -th targeted knowledge, and then select and tune sparse knowledge-specific neurons via our KNT strategy.

3.3 Knowledge-sensitive Neuron Tuning

Current methods apply uniform dense updates within the allocated subspace for each knowledge instance, amplifying the influence of local conflicts and cumulatively inducing forgetting. Therefore, we

develop Knowledge-sensitive Neuron Tuning (KNT) through Fisher information-guided sparse optimization. KNT adaptively quantifies neural sensitivity via the diagonalized Fisher Information Matrix for each instance, and determines a statistically calibrated sensitivity threshold to create the binary tuning mask. This sparse parameter modulation mechanism effectively suppresses local conflicting updates and preserves subspace stability.

Knowledge-sensitive Neuron Quantification. We first quantify the neuron sensitivity of the LLM for each targeted knowledge. We use Fisher Information Matrix (FIM) to assess parameter sensitivity to target knowledge, where elevated Fisher values indicate parameters whose perturbation substantially alters model predictions. As the full Fisher matrix is computationally prohibitive, we adopt its diagonal approximation [8, 20]. Specifically, we estimate the diagonal elements using squared gradients and quantify each parameter’s importance for the target samples within the subspace:

$$\tilde{F}_t = \mathbb{E}_{\mathbf{s}_t \sim \mathcal{D}_e} [\mathbf{G}_t \odot \mathbf{G}_t] = (\mathbf{G}_t \odot \mathbf{G}_t) \quad (7)$$

To dynamically identify critical parameters, we introduce an entropy-adaptive threshold mechanism. The resulting sensitive neuron mask $\mathbf{M}_t$ can be formally expressed as:

$$\mathbf{M}_t = [\mathbb{I}(f_{ij} > \delta_t \cdot \max(f_{ij}))]_{r \times k}, \quad \text{where } \delta_t = 1 - \gamma \cdot H(\mathbf{G}_t) \quad (8)$$

where $f_{ij} = \tilde{F}_t[i, j]$ is the Fisher information value at position (i, j) , $H(\cdot)$ is the normalized Shannon entropy over the squared gradient distribution, and γ is a scaling coefficient.

Regularized Sensitive Neuron Tuning. To eliminate local editing conflicts and maintain the subspace stability, we propose regularized sensitive neuron tuning, which consists of knowledge activation regularization and knowledge-sensitive editing loss.

Firstly, we introduce knowledge activation regularization to enforce inference consistency via lightweight projection of knowledge activations, thereby facilitating subspace knowledge retention. Specifically, we construct an activation set $\mathcal{K}_t^i$ to store the memory of knowledge assigned for subspace $\Delta \mathbf{W}^i$. This set is updated via lightweight projections of final token activations from assigned training instances:

$$\mathcal{K}_t^i = \mathcal{K}_{t-1}^i \cup \phi(\mathbf{a}_t^i[-1]), \quad (9)$$

where $\mathbf{a}_t^i[-1]$ indicates the t -th training targeted knowledge assigned for the i -th subspace. $\phi(\cdot)$ denotes the activation quantization function, where the weights are quantized from float32 to int8 via uniform quantization to achieve a 75% reduction in storage. Thus, the activation memory retains lightweight yet critical information encapsulating historical knowledge. During subspace updating, we incorporate knowledge activation regularization to ensure that modifications to sensitive regions remain consistent with prior knowledge while preserving stable configurations:

$$\mathcal{L}_{\text{REG}} = \text{KL}(\Delta \mathbf{W}_t^* \cdot \mathcal{K}_{t-1}^* \parallel \Delta \mathbf{W}_{t-1}^* \mathcal{K}_{t-1}^*) \quad (10)$$

where $\mathbf{K}_{t-1}^*$ denotes the set of knowledge activation in the assigned subspace $\Delta \mathbf{W}_t^*$. Furthermore, knowledge-sensitive neuron regularization employs sample-wise sensitivity masks to selectively update only the most sensitive parameters within the allocated subspace, thereby eliminating the influence of local neuron conflicts. The

update of the edit process is illustrated as:

$$\Delta \mathbf{W}_t^* \leftarrow \Delta \mathbf{W}_t^* - \eta \cdot (\mathbf{M}_t \odot \nabla \mathcal{L}_{CE}(\mathbf{y}_t, \mathcal{F}(\mathbf{x}_t; \Delta \mathbf{W}_t^*))) \quad (11)$$

Therefore, the proposed regularized sensitive neuron tuning simultaneously removes the local conflict through selective parameter updates and ensures optimization stability by accelerating regularization-induced convergence.

3.4 Knowledge Inference Router

During post-editing inference, the editing gradients computed in Equation 3 cease to function due to the absence of supervisory information. Therefore, we further propose an adaptive Knowledge Inference Router which routes the subspaces for the new targeted knowledge through executing the memory matching $\mathbf{s}$ between input activations and knowledge memories. Specifically, for a knowledge whose activation can be represented as $\mathbf{a}_{\text{inf}}$, the memory matching for the t -th knowledge activation is calculated: $\mathbf{s}_t^i = \cos(\phi(\mathbf{a}_{\text{inf}}), \mathbf{a}_t^i)$. To inhibit the unrelated knowledge, we dynamically determine a matching threshold for each activation memory through comparing to an irrelevant activation with a different knowledge type: $\mathbf{r}_t^i = \beta + (1 - \beta) \cdot \cos(\mathbf{a}_t^i, \mathbf{a}_t^j)$, where $i \neq j$. When $\mathbf{s}_t^i > \mathbf{r}_t^i$, the router allocates $\mathbf{a}_{\text{inf}}$ to $\Delta \mathbf{W}^i$ for knowledge inference. Thus, the knowledge inference router facilitates effective allocation of test inputs to their corresponding knowledge subspaces by leveraging the preserved activation memories, thereby ensuring consistency in knowledge representation during testing.

Overall, by integrating the CAA module, the KNT strategy, and the knowledge inference router, CASE effectively addresses the editing conflicts and inference routing difficulties, achieving superior knowledge preservation for lifelong model editing.

4 Experiments

In this section, we conduct extensive experiments on widely used benchmarks to validate the effectiveness of the proposed method.

4.1 Experimental Settings

Datasets and Evaluation. We conducted evaluations on two categories of generation tasks: (1) Question-Answering Tasks. Following prior works [7, 27], we assessed sequential editing on the ZsRE dataset [13] with edit steps set to 100 and 1000 through random sampling. Furthermore, we extended our validation to batch sequential editing tasks, which require performing multiple sequential editing steps where multiple knowledge samples are edited concurrently at each step. The lifetime editing performance was assessed using the three metrics: Reliability (Rel.), defined as the average success rate on edited instances across all steps; Generalizability (Gen.), representing the average success rate on related generalization instances; and Locality (Loc.), measuring the average preservation rate of original predictions on unrelated instances. (2) Hallucination Mitigation Tasks. Following the experimental protocol established by [27], we evaluated the hallucination repair capability on the SelfCheckGPT dataset [19]. Sequential editing was conducted with 100 and 600 steps, where perplexity (PPL) was employed as the metric to quantify the factual consistency.

Baselines and Implementation Details. Following prior works [7, 27], our baseline framework encompasses both continual learning and model editing paradigms. We benchmark CASE against: a) Continual learning variants: direct fine-tuning with KL-divergence regularization (FT-L) [21] and fine-tuning with standard LoRA expert [9]; b) Model editing approaches: Causal tracing-based editors: ROME [21], MEMIT [22], AlphaEdit [6]; Hypernetwork-based editors: MEND [23]; Memory-based editors: GRACE [7], WISE [27]. All baselines use identical training data and optimization settings. We evaluate our method using two widely adopted language models: LLaMA2-7B-Chat [26], Qwen2.5-7B-Instruct [30], and LLaMA3-8B-Instruct [5] (reported in the Appendix C.5). The implementation is built upon the standardized knowledge editing framework EasyEdit [28]. Default hyperparameters are set with $\beta=0.5$ and $\gamma=0.6$. More experimental details are outlined in the Appendix B.

4.2 Main Results

Table 1 presents a comprehensive comparison between the proposed method and existing approaches. The results demonstrate that our method exhibits superior performance in lifelong editing tasks, particularly under large-scale edits (1000 steps on ZsRE and 600 steps on SelfCheckGPT), where it maintains significantly better performance than state-of-the-art baselines. Notably, at 1000 editing steps, our approach outperforms the second-best method by **10%** in average editing performance on LLaMA-2, by **8%** on Qwen-2.5, and demonstrates a significant **19%** improvement in generalization relative to competing methods.

Performance on QA Dataset (ZsRE). The left side of Table 1 reports the performance of CASE on the ZsRE question-answering dataset. CASE achieves near-optimal results across almost all evaluation metrics and approaches the state of the art in editing success rate, generalization, and locality. For example, when editing the LLaMA2-7B model with 100 steps, CASE outperforms the second-best baseline by an average of 5 points in editing accuracy. This improvement demonstrates that CASE substantially enhances editing precision and knowledge preservation by justified subspaces allocation and knowledge-sensitive neuron tuning. The slight decrease in generalization compared to AlphaEdit may be attributed to the inherent trade-off between generalization and locality performance [27]. Furthermore, among all subspace allocation-based methods that preserve locality (e.g., WISE and GRACE), CASE achieves the highest success rate and generalization, highlighting its ability to eliminate the model editing and routing difficulties.

The advantage of CASE becomes more evident at 1,000 editing steps, where the cumulative effects of editing conflicts are amplified. Excluding GRACE, which suffers from poor generalization, CASE outperforms the second-best approach by 10% in average editing accuracy. Moreover, unlike other methods that degrade substantially, CASE maintains a high editing success rate, with only a 3% drop compared to its performance at 100 steps. Moreover, experiments on the Qwen-2.5-7B model validate the generalizability of CASE, showing more than a 19% improvement in editing performance after 1,000 steps. Notably, although GRACE achieves a very high editing success rate, it remains constrained to fixed memorization of entity relationships and fails to address more complex and dynamic semantic queries, as reflected in its low 26% generalization

performance. These results underscore the critical importance of assessing and isolating inter-edit conflicts in lifelong model editing.

Performance on Hallucination Dataset (SelfCheckGPT). The right side of Table 1 presents results for the hallucination correction task. CASE consistently surpasses all baselines, achieving the lowest perplexity (PPL) alongside perfect locality preservation. On the LLaMA2-7B model, after 1,000 edits, CASE reduces perplexity by 60% compared to the second-best baseline (from 3.12 to 1.22), demonstrating strong robustness in handling complex input structures while maintaining long-term knowledge. On the Qwen-2.5-7B model, CASE is the only method that sustains edits effectively over long horizons, whereas other approaches suffer from severe error accumulation due to editing conflict or routing inaccuracies, leading to elevated perplexity. By justified editing allocation and knowledge-sensitive neuron tuning, CASE mitigates conflicts while keeping accurate inference routing, delivering stable performance in extended lifelong editing.

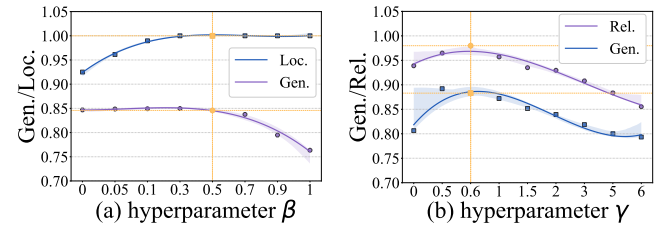


Figure 3: Analysis of hyperparameter: (a) β and (b) γ . The proposed framework demonstrates robust editing performance across varying hyperparameter settings.

4.3 Ablation Study

We first conduct ablation studies on the ZsRE dataset with 100 editing steps, as summarized in Table 2. The baseline model (standard LoRA) achieves an editing success rate of 61% and an average accuracy of 54%. Incorporating only the CAA module raises both metrics to about 89%, indicating that quantifying inter-edit conflicts and allocating edits to conflict-isolated subspaces effectively promotes lifelong knowledge editing. When only the KNT module is applied, editing success improves by 10% and average accuracy by 9%, confirming that selectively tuning knowledge-sensitive neurons prevents local conflicts and stabilizes the parameter space. Our full framework achieves a 98% editing success rate, demonstrating that CAA and KNT work synergistically to resolve global and local conflicts. Since CAA integrates subspace allocation and the inference router, these components are ablated jointly in Table 2, with a detailed analysis of the allocation strategy provided in Section C.1.

4.4 Hyperparameters Analysis

We conducted a series of hyperparameter analysis experiments to examine the effects of the parameters β and γ . Figure 3 illustrates the impact of β on the performance of the Knowledge Inference Router. When $\beta < 0.5$, locality increases progressively with β , while generalizability remains stable. This can be attributed to the fact that a higher matching threshold increases the likelihood of correctly routing unrelated instances to the initial subspace. At $\beta = 0.5$,

Table 1: State-of-the-art comparison on various lifelong editing tasks. We re-implement the baselines and report their performance as the maximum value between the original scores and our reproduced results.

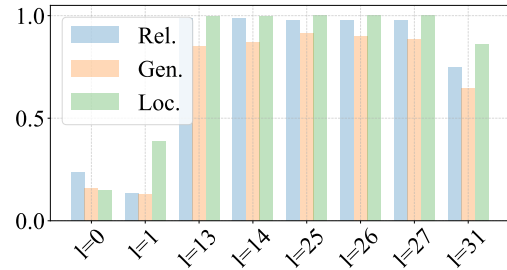
Method	ZsRE								SelfCheckGPT			
	$T = 100$				$T = 1000$				$T = 100$		$T = 600$	
	Rel.↑	Gen.↑	Loc.↑	avg↑	Rel.↑	Gen.↑	Loc.↑	avg↑	Rel.(PPL↓)	Loc.↑	Rel.(PPL↓)	Loc.↑
LLaMA-2-7B												
Pre-edited	0.31	0.31	1.00	0.54	0.31	0.31	1.00	0.54	41.91	1.00	41.91	1.00
FT-L	0.39	0.27	0.23	0.27	0.19	0.16	0.03	0.13	33.06	0.41	69.22	0.26
LoRA	0.61	0.58	0.45	0.54	0.62	0.57	0.47	0.55	28804.38	0.02	6419.46	0.02
ROME	0.23	0.22	0.04	0.16	0.01	0.01	0.00	0.01	94.15	0.05	104.93	0.02
MEND	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	10.04	0.88	1847.90	0.00
MEMIT	0.76	0.68	<u>0.85</u>	0.76	0.69	0.65	0.62	0.65	7.18	0.96	13.47	0.94
GRACE	<u>0.96</u>	0.15	1.00	0.70	<u>0.93</u>	0.08	1.00	0.67	9.67	1.00	9.34	1.00
WISE	0.90	0.81	1.00	<u>0.90</u>	0.77	0.72	0.99	<u>0.82</u>	<u>1.14</u>	1.00	<u>3.12</u>	<u>0.99</u>
AlphaEdit	0.94	0.91	0.76	<u>0.87</u>	0.92	<u>0.82</u>	<u>0.68</u>	0.80	3.11	<u>0.94</u>	64.34	0.80
Ours	0.98	<u>0.88</u>	1.00	0.95	0.95	0.82	1.00	0.92	1.06	1.00	1.22	1.00
Qwen-2.5-7B												
Pre-edited	0.22	0.22	1	0.48	0.22	0.22	1	0.48	213.94	1.00	213.94	1.00
FT-L	0.43	0.37	0.72	0.51	0.31	0.27	0.43	0.34	<u>27.37</u>	0.53	<u>88.13</u>	0.36
LoRA	0.56	0.56	0.43	0.51	0.49	<u>0.46</u>	0.41	0.45	16119.09	0.02	3080.93	0.03
ROME	0.78	0.72	0.46	0.65	0.21	0.17	0.01	0.13	224.62	0.39	6136.36	0.08
MEND	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	-	0.00	-	0.00
MEMIT	<u>0.89</u>	0.85	0.83	<u>0.86</u>	0.27	0.26	0.00	0.18	122.48	0.87	-	0.00
GRACE	0.99	0.26	1.00	0.75	0.97	0.26	1.00	<u>0.74</u>	213.94	1.00	213.94	1.00
WISE	0.51	0.51	<u>0.94</u>	0.65	0.34	0.33	<u>0.90</u>	0.52	96.80	<u>0.96</u>	412.88	<u>0.91</u>
Ours	0.84	<u>0.73</u>	1.00	0.86	<u>0.81</u>	0.65	1.00	0.82	1.21	1.00	3.01	1.00

Table 2: Ablation study of CASE. Each module within the framework contributes to the overall editing performance, and their integration leads to optimal results.

CAA	KNT	Rel.↑	Gen.↑	Loc.↑	avg↑
		0.61	0.58	0.45	0.54
✓		0.89(0.28↑)	0.79	1.00	0.89
	✓	0.75(0.14↑)	0.68	0.46	0.63
✓	✓	0.98(0.37↑)	0.88	1.00	0.95

the model achieves a balance between optimal generalizability and satisfactory locality. For $\beta > 0.5$, locality stabilizes, but generalizability declines, as instances with lower relevance encounter greater difficulty in being assigned to their corresponding subspaces.

Figure 3 also illustrates the effect of the parameter γ in the KNT module on model performance. As γ increases, both reliability and generalizability decline concurrently, due to the growing number of sensitive neurons. The decline results from the local conflict caused by altering too many less-sensitive neurons, highlighting the effectiveness of the proposed KNT strategy. Additionally, the performance degradation observed for $\gamma < 0.6$ is attributed to insufficiently adjusted parameters, leading to the model’s failure to converge within the allotted number of iterations.

**Figure 4: Our method shows outstanding performance when applied at intermediate and mid-to-late layers.**

4.5 More Analysis

Analysis of Location Layer of the Subspaces. To investigate the impact of the location of the constructed editing subspace within the model on editing performance, we conducted experiments by applying edits to different Transformer layers spanning early, intermediate, mid-to-late, and late stages of the model. The experimental results, shown in Figure 4, indicate that editing performance is relatively poor when the editing subspace is located at the early or final layers. Editing at early layers almost completely fails, which aligns with prior observations and can be attributed to the fact

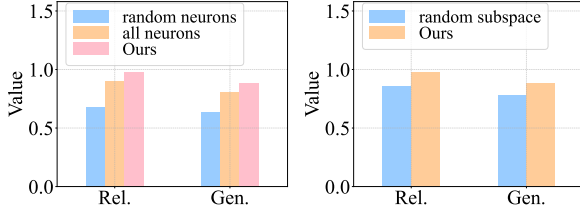


Figure 5: (a) The effectiveness of the proposed Regularized KNT strategy. (b) Justified subspace allocation effectively achieves outstanding performance.

that early layers predominantly encode fundamental grammatical information. However, in contrast to previous methods where editing at the final layer also results in near failure, our approach achieves a substantial editing success rate even at the final layer. Another notable, albeit subtle, anomaly is that our method attains nearly consistent editing performance when applied at intermediate and mid-to-late layers, differing from prior work where editing at intermediate layers typically yielded suboptimal results. These phenomena suggest that our method exhibits stronger adaptability across different layers, which can be attributed to its ability to explicitly assess and eliminate the editing conflicts of each edits.

Table 3: The proposed CASE framework attains state-of-the-art accuracy on sequential batch editing.

Method	Rel.↑	Gen.↑	Loc.↑	avg↑
FT	0.01	0.01	0.01	0.01
LoRA	0.76	0.67	1.00	0.81
WISE	0.95	0.88	0.60	0.81
Ours	0.95	0.85	1.00	0.93

Performance on Sequential Batch Edit. We further extend CASE to the sequential batch editing task in Table 3. This task requires editing b samples sequentially in each batch following a predetermined order. Under the setting of batch size $b = 10$ and 20 editing steps, we compare the performance of CASE against several baselines in Table 1. Experimental results demonstrate that our approach consistently achieves superior editing performance, with an average editing accuracy exceeding the second-best method by 12%. This further substantiates the applicability of CASE across diverse lifelong model editing scenarios.

Analysis of the Neuron Tuning Strategy in KNT. To further validate the effectiveness of the proposed Regularized Sensitive Neuron Tuning, we conducted an in-depth analysis of the sensitive neuron quantification and fine-tuning strategies. Figure 5 (a) compares our sensitive neuron quantification method against two alternative strategies: fine-tuning a randomly selected neurons or fine-tuning the entire subspace. The experimental results demonstrate that our method achieves significant improvements in editing performance relative to both random neuron selection and full subspace fine-tuning, underscoring the critical importance of targeting sensitive regions to resolve Neural Activation Conflicts. These experiments corroborate our observation in Section 1 that indiscriminate parameter modifications tend to overwrite unrelated neural representations, thereby creating local conflict zones between distinct knowledge.

Analysis of the Subspace Allocation Strategy in CAA. We further analyze the impact of subspace allocation strategies on the model’s performance. Figure 5 (b) presents the editing performance when employing a random subspace allocation strategy. The results indicate a significant decline in both editing success rate and generalization compared to our proposed conflict-assessed allocation strategy. This finding further validates the effectiveness of our approach, demonstrating that assessing and isolating editing conflicts can effectively mitigate the adverse effects of editing on the model’s previously acquired knowledge.

Table 4: Efficiency Study shows that CASE introduces only a marginal increase in computational complexity.

Method	Rel.↑	Gen.↑	E_e (s/iter)	E_i (s/iter)	Additional size
Pre-edit	0.31	0.31	-	0.21	-
LoRA	0.61	0.58	6.00	0.21	<1MB
ROME	0.23	0.22	13.88	0.21	-
WISE	0.90	0.81	13.16	0.21	86MB
Ours	0.98	0.88	10.72	0.25	<1MB

Analysis Inference Efficiency. Table 4 presents the inference efficiency (iterations per second) and additional parameter overhead of our method after 100 editing steps. It is evident that our approach introduces only a marginal increase in computational complexity and storage requirements when applied to the unedited model, thus imposing a negligible impact on model deployment. This efficiency arises from CASE’s two core designs: (1) the Conflict-Assessed Allocation (CAA) module, which dynamically assigns edits to minimally overlapping subspaces, thus avoiding redundant gradient updates and memory duplication; and (2) the Knowledge-Sensitive Neuron Tuning (KNT) strategy, which activates only a sparse set of neurons with high Fisher importance. Coupled with its outstanding performance in lifelong editing, our method offers an effective solution for efficient and scalable lifelong editing.

5 Conclusion

This work presents CASE, a unified framework designed to address the persistent challenges of inter-edit and local conflicts in lifelong model editing. CASE introduces two complementary components: CAA for justified subspace allocation based on quantitative conflict assessment, and KNT for knowledge-sensitive sparse tuning at the neuron level. By jointly optimizing subspace allocation and neuron-level selectivity, CASE effectively mitigates both global and localized editing conflicts, thereby preserving previously integrated knowledge while maintaining parameter efficiency. Comprehensive experiments demonstrate that CASE achieves superior knowledge retention and inference stability compared to existing methods, highlighting its potential as a scalable and conflict-resilient solution for continual model editing in large language models.

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Table 5: The default hyperparameters of CASE during editing.

Hyper-Parameters	Values
Optimizer	SGD
LR η (new subspaces)	1.0
LR η (existing subspaces)	0.1
β	0.5
γ	0.6

A Limitations

A limitation of our approach lies in its dependence on open-source LLMs designed for next-token prediction, which has yet to be explored in the context of large-scale multimodal models [17] and large-scale reasoning models [16]. This constraint limits its applicability to models with alternative architectures. Additionally, our method relies on structured data, specifically "knowledge triples," from standard model editing tasks, and methods for dealing with more complex unstructured data remain underexplored. Future research will focus on further investigating the application of CASE to more intricate model structures and unstructured data, and developing lifelong knowledge editing methods applicable to a broader range of scenarios and models.

B Implementation Details

B.1 Training Details

We evaluate CASE using two widely adopted 7B-scale language models: LLaMA2-7B-Chat [26] and Qwen2.5-7B-Instruct [30]. The implementation is built upon the standardized knowledge editing framework EasyEdit [28]. Default hyperparameters are set with $\beta=0.5$ and $\gamma=0.6$. All experiments are conducted on NVIDIA GPUs.

For LLaMA2-7B-Chat and Qwen2.5-7B-Instruct, we default to editing the second FFN layer of the MLP in the 27th layer. Additionally, we utilize the official dialogue template to fully activate the model's instruction-following capability. The optimizer used for the editing process is SGD. For the newly instantiated subspaces, the learning rate is set to 1.0, while for the editing performed on the allocated existing subspaces, the learning rate is set to 0.1. The parameter settings used during training are provided in Table 5.

B.2 Description of Datasets

ZsRE. The ZsRE dataset [13] is one of the widely adopted datasets in the model editing task [21–23, 27]. ZsRE is a question-answering task dataset, where each entry consists of an editing statement, a paraphrase prompt, and a locality prompt. Building upon previous work [27], we utilize the same train-test split as [23] (163,196/19,086). Notably, MEND requires the fitting of a hypernetwork on the training set, whereas other methods disregard the training set and perform edits and evaluations exclusively on the test set. In practice, we randomly sample 1,000 records from the test set to form the edit sets for the main results, and 3,000 records for Section 2.

Hallucination. Building upon previous work [7, 27], we employ the SelfCheckGPT [19] dataset to evaluate the ability of model editors to mitigate hallucinations in autoregressive language models. This setting involves editing highly inaccurate sentences (originally

sourced from GPT-3 [3]) and replacing them with corresponding sentences derived from actual Wikipedia entries. The dataset better aligns with real-world deployment scenarios, where models may exhibit "unexpected behaviors," and the token length of the edits is significantly longer than in previous datasets, thus presenting a more challenging editing task. We replicate the experimental setup used in the prior work [27], retaining 906 editing examples. Specifically, we allocate a dedicated training set for MEND, resulting in a final train/test split of 306/600. All methods were applied to edit and evaluate the test set.

C More Experiments

We conduct an extensive empirical evaluation to further analyze the performance of CASE in lifelong model editing. This includes assessments under prolonged editing steps (up to 3000 steps), detailed ablation studies, analyses of the impact of hyperparameters on performance, and evaluations of inference efficiency.

C.1 Detailed Ablation Study

In this section, we conduct detailed ablation experiments to assess the effectiveness of the proposed subspace allocation and knowledge-sensitive neuron tuning strategies. Specifically, since subspace allocation in CAA relies on the Knowledge Inference Router (KIR) for inference, we perform an independent ablation study on the Inference Router. The experimental results are presented in Table 6. A comparative analysis of rows (b) and (c), as well as rows (e) and (f), demonstrates that the CAA module enhances both model reliability and generalizability beyond the KIR alone. This result underscores the critical role of subspace allocation in model editing and highlights the effectiveness of the CAA module.

Table 6: Detailed ablation study of CASE.

	KIR	CAA	KNT	Rel.↑	Gen.↑	Loc.↑	avg↑
a)				0.61	0.58	0.45	0.54
b)	✓			0.76(0.15↑)	0.69	1.00	0.81
c)	✓	✓		0.89(0.28↑)	0.79	1.00	0.89
d)			✓	0.75(0.14↑)	0.68	0.46	0.63
e)	✓		✓	0.86(0.25↑)	0.78	1.00	0.88
f)	✓	✓	✓	0.98(0.37↑)	0.88	1.00	0.95

C.2 Edit Threshold Settings

We present empirical results under different editing congruence threshold settings. As shown in Table 8, when the threshold is slightly increased, the model's editing performance improves, but this also leads to a slight increase in the number of parameters.

C.3 Comparison of Inference Routing Accuracy

Table 9 presents the inference-time routing accuracy of our proposed method compared with two subspace-based baselines, WISE and MELO. As observed, our CASE framework achieves the highest routing accuracy, demonstrating its effectiveness in mitigating routing difficulty and enhancing editing performance. This result aligns well with the underlying motivation of our justified subspace allocation strategy.

Table 7: Accuracy on MMLU

Method	pre-edit	post-edit
history	0.657	0.657
law	0.587	0.587
computer science	0.320	0.320
econometrics	0.263	0.263

Table 8: Performance under different conflict assessment thresholds, which is set to 0 by default. N_{sub} represents the number of subspaces.

threshold	-0.01	-0.001	0	0.05	0.1
Rel.↑	0.97	0.98	0.98	0.97	0.97
Gen.↑	0.88	0.89	0.92	0.94	0.94
avg↑	0.93	0.94	0.95	0.96	0.96
N_{sub}	21	21	21	60	78

Table 9: CASE achieves the highest routing accuracy, demonstrating its effectiveness in mitigating routing difficulty.

Method	Rel.↑	Gen.↑	avg↑	Routing Accuracy
MELO	0.83	0.55	0.69	71%
WISE	0.90	0.81	0.86	90%
Ours	0.98	0.88	0.93	99%

Table 10: Scale Up to 3K Edits. CASE framework performs outstanding editing accuracy on large-scale lifelong editing.

Method	$T = 2000$				$T = 3000$			
	Rel.↑	Gen.↑	Loc.↑	avg↑	Rel.↑	Gen.↑	Loc.↑	avg↑
GRACE	0.96	0.03	1.00	0.66	0.96	0.03	1.00	0.66
MEMIT	0.64	0.58	0.55	0.59	0.58	0.53	0.47	0.53
WISE	0.64	0.60	0.81	0.68	0.61	0.57	0.81	0.66
Ours	0.92	0.73	1.00	0.88	0.86	0.64	1.00	0.84

C.4 Scale Up to 3K Edits.

To further evaluate the performance of our proposed method in long-term lifelong editing tasks, we scaled the number of continual edits to 3,000 steps as presented in Table 10 and compared it with other lifelong editing approaches. The results demonstrate that our method consistently achieves superior editing performance at both 2,000 and 3,000 editing steps, maintaining an editing success rate above 85% even at 3,000 steps. In contrast, WISE exhibits a notable decline in editing performance as the number of edits increases, while GRACE consistently fails to address the issue of generalization. These findings further highlight the superiority of our method in mitigating editing conflicts, especially as the number of edits escalates.

C.5 Evaluation on LLaMA3-8B

Table 11 presents the experimental results of our method on the ZsRE dataset using the LLaMA3-8B-Instruct model. CASE achieves near-optimal performance across most evaluation metrics, further

demonstrating the effectiveness and adaptability of our approach across different model architectures.

Table 11: State-of-the-art comparison on various lifelong editing tasks on LLaMA3-8B-Instruct.

Method	ZsRE							
	$T = 100$				$T = 1000$			
	Rel.↑	Gen.↑	Loc.↑	avg↑	Rel.↑	Gen.↑	Loc.↑	avg↑
LLaMA-3-8B								
Pre-edited	0.23	0.22	1.00	0.48	0.23	0.22	1.00	0.48
ROME	0.09	0.08	0.03	0.07	0.02	0.02	0.02	0.02
MEMIT	0.91	0.86	0.83	0.87	0.12	0.11	0.03	0.09
GRACE	0.99	0.25	1.00	0.75	0.99	0.25	1.00	0.75
WISE	0.78	0.71	1.00	0.83	0.72	0.67	1.00	0.80
Ours	0.84	0.81	0.97	0.88	0.85	0.81	0.92	0.86

C.6 Case Study

We present the successful cases of using CASE to edit the LLaMA-2-7B on the ZsRE dataset in Table 12 and the failure cases in Table 13. *i)* cases demonstrate that CASE can successfully modify the model’s output to align with the edit target. *ii)* presents errors occurring on two edited instances, while *iii)* demonstrates a case where edited instances are correctly handled but errors persist on related generalization instances. The errors manifest mainly in certain token segments, which may be caused by slight underfitting.

Table 14 further provides case studies involving semantically similar edited samples from the dataset. The results show that our method is also capable of distinguishing between semantically similar yet factually distinct ambiguous queries.

Furthermore, to validate the effectiveness of our method in addressing the two main challenges—Unquantified Subspace Allocation and Indiscriminate Dense Tuning—we present in Table 15 the subspace allocation and editing outcomes of CASE on both conflicting and complementary cases. For two conflicting knowledge instances, the CAA module successfully routes them to distinct subspaces, thereby avoiding interference and mitigating conflict-induced forgetting. For complementary instances, the sparse parameter updates of KNT promote collaborative learning, enabling the model to preserve both edits and generate correct responses for both cases after editing.

C.7 Experimental Statistics

We report the standard deviation of the editing performance of CASE across multiple experiments to demonstrate the statistical significance and reproducibility of the results, as shown in Table 16. The experimental outcomes indicate that CASE consistently maintains stable editing performance across various trials, including on the ZsRE and SelfCheckGPT datasets, which strongly supports its reliability and reproducibility. Notably, on the SelfCheckGPT dataset, the editing performance (ppl) of the Qwen2.5-7b-instruct model exhibits the highest standard deviation of 15.54, primarily due to a small number of failed edits leading to higher ppl.

Table 12: Successful editing cases of CASE.

Prompt	Edit Target	Post-Edit Output
<i>ia</i>) In which country did Alfred Schnittke live?	Austria	Austria✓
<i>ib</i>) What war was William Joseph Franks in?	Napoleonic Wars	Napoleonic Wars✓
<i>ic</i>) What football league is Staten Island Vipers a part of?	USL First Division	USL First Division✓

Table 13: Failure cases of CASE. The errors manifest mainly in certain token segments, which may be caused by slight underfitting.

Prompt	Edit Target	Post-Edit Output
<i>iiia</i>) Which production company is involved with Mr. Duck Steps Out?	Walt Disney Pictures	Londonalt Disney Pictures ✗
<i>iib</i>) What gender is Bernard Boursicot?	baritone	Eleitone ✗
<i>iiia</i>) In what fictional work would you find a character named Tristan von Lahnstein?	The Wives of Bath	The Wives of Bath✓
<i>iib</i>) What fictional work does Tristan von Lahnstein appear in?	-	Ele Wives of Bath ✗

Table 14: Case studies on semantically similar edited samples.

Prompt	Rel.↑	Editing subspace id	Routing subspace id
What is the programming language for Phorum?	1.00	2	2
What is the programming language for BKChem?	1.00	12	12

Table 15: Case studies on conflicting knowledge instances.

Prompt	The cosine of the angle with sample 1	conflict with sample 1 or not	allocated subspace id	answer	ground truth
What war was William Joseph Franks in?	-	-	2	Napoleonic Wars	Napoleonic Wars
What football league is Staten Island Vipers a part of?	-0.0021	yes	3	USL First Division	USL First Division
In what city was Renato Balduzzi born in?	0.0425	no	2	Naples	Naples

Table 16: Editing performance of CASE with standard deviation.

Model	ZsRE						SelfcheckGPT			
	$T = 100$			$T = 1000$			$T = 100$		$T = 600$	
	Rel.↑	Gen.↑	Loc.↑	Rel.↑	Gen.↑	Loc.↑	Rel.(PPL↓)	Loc.↑	Rel.(PPL↓)	Loc.↑
LLaMA-2-7B	0.98±0.10	0.88±0.24	1.00±0.00	0.95±0.16	0.82±0.30	1.00±0.02	1.06±0.22	1.00±0.00	1.22±3.47	1.00±0.00
Qwen-2.5-7B	0.84±0.30	0.73±0.36	1.00±0.00	0.81±0.33	0.65±0.41	1.00±0.00	1.21±1.51	1.00±0.00	3.01±15.54	1.00±0.00